

Interaction of Learner Control and Prior Understanding in Computer-Assisted Video Instruction

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It is commonly assumed that individual differences in abilities and aptitudes are accommodated if learners have more control over pace, amount of practice, or style of instruction they receive. In this study, students with varying degrees of prior conceptual understanding of a scientific topic were randomly assigned to a program-controlled or learner-controlled treatment in a sequence of computer-assisted video learning. A 2×2 analysis of variance was conducted on posttest scores and time on task. As expected, low-aptitude subjects in the learner-controlled treatment had significantly lower posttest scores than high-aptitude subjects in both the program-controlled and learner-controlled treatments and low-aptitude subjects in the program-controlled treatment. High-aptitude subjects were significantly more efficient in their use of time in the learner-controlled treatment than subjects in the other three treatments.

Computer-assisted instruction using interactive video-discs is a new technology that combines the power of the microcomputer with the video and audio capabilities provided by the laser videodisc. Interactive videodisc systems provide random access to a variety of video, audio, and computer information and allow a high degree of learner interaction.

According to Salomon (1979), "The technology of medium affects the knowledge acquired" (p. 8). Despite the importance of the medium's effect on cognition, little is understood about how various attributes of computer-assisted video affect learning outcomes. This is an important issue because interactive video systems provide vast amounts of information resources in new forms. As the amount of information increases, so also does the number of ways it can be put together and accessed by learners. The ways in which learners access information and make decisions about which information to retrieve may affect their ability to assimilate, retain, and use it later on.

One of the benefits of computer-based learning systems is the control that can be given to individuals. Learners may control the pace, sequence, depth of study, or style of instruction. There is some evidence that increased control enhances feelings of self-efficacy and self-determination and helps learners take independent responsibility for their own learning and behavior (Landa, 1976; Papert, 1980). Learner-controlled methods can accommodate individual differences in initial aptitude (Bunderson, 1980; Merrill, 1979). The results from experimental studies, however, have been contradictory (Judd, 1972; Steinberg, 1977). In most studies, students have actually learned less when they have had

control over their own instructional strategies. The few studies that have found positive advantages for learner control have involved highly motivated and/or intelligent subjects, who might be expected to do better under less structured conditions (Judd, 1975; Tennyson & Rothen, 1979).

Learner control cannot overcome the fact that "individual characteristics not under control of the individual will determine to a significant extent what and how much that individual will learn in a given instructional setting" (Snow, 1980, p. 1). Aptitude-treatment interaction (ATI) proposes that different instructional approaches are appropriate for students of different aptitudes. Generally, ATI researchers test the hypothesis that the higher the level of prior familiarity, the lower the program support needed to accomplish the objectives (Cronbach & Snow, 1977; Ross, 1984). Conversely, as the level of prior familiarity decreases, the amount of required program support increases.

The major focus of this study was to examine the effects of prior conceptual understanding and learner control on performance. The general hypothesis was that subjects who had some prior conceptual understanding would learn best when given control (learner control) and that subjects with lower prior conceptual understanding would learn best when given some structure (program control).

Method

Subjects

Students from Introductory Biology 103 at Cornell University participated in the study. A 23-item pretest was administered to 156 volunteer subjects. The test (Cronbach alpha = .73) measured an individual's understanding of the protein synthesis process. On the basis of pretest scores, we randomly selected 80 subjects, 40 from the low-conceptual-understanding group (raw scores between 3 and 8) and 40 from the high-conceptual-understanding group (raw scores between 11 and 17). Scholastic Aptitude Test scores revealed that there were no significant differences in aptitude between groups.

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Table 1
Major Features of the Two Treatment Conditions

Program-controlled	Learner-controlled
Student control of pace	Student control of pace
Computer-controlled presentation of concepts in hierarchical form	Student control of sequence
Student's progress diagnosed automatically and remediation and review provided as needed	Student control of depth or amount of practice
	Student choice of mode of presentation (video, audio, graphics, or text)
	Student choice of type of content (rules, key ideas, examples, or practice)

Design

The study used a 2 (program control vs. learner control) $\times$ 2 (low conceptual understanding vs. high conceptual understanding) factorial design. Subjects were assigned to receive the program- or learner-controlled treatment by using systematic randomization to obtain equal numbers in each treatment condition (Kachigan, 1982). The dependent variables were posttest scores and time on task.

Materials

The hardware consisted of an IBM personal computer connected to a video player through a video-controlled interface board. Video images were taken from the videodisc entitled *The Development of Living Things* and developed by WICAT Systems. Courseware was designed and programmed by the researcher to meet the particular requirements of the experiment.

Procedure

Subjects were allotted a 2-hr time period in which to study four modules on DNA structure and the translation and transcription phases of protein synthesis. Two treatments were administered.

First, under the program-controlled treatment, material was sequenced logically by ordering the computer text, graphics, and video and audio segments to reflect the hierarchical nature of the topic structure. Subjects had to complete the learning task in a

specified order. The objectives of the lesson were presented at the beginning of each module. A typical module consisted of the rule, example, and practice format. Video and audio segments were used to supplement computer-generated text and graphics. An individual's progress was automatically diagnosed by the computer, and remediation and feedback were provided as needed. Subjects controlled only the pace of the lessons.

Second, under the learner-controlled treatment, subjects controlled not only the pace of the lessons but also the sequencing, presentation mode (video, audio, graphics, or text), amount of practice, and depth of study. Subjects were able to select the order of lessons and modules and the number of examples and practice items, and they could elect to receive remediation or review. Subjects accessed the learner-controlled options through menus or computer function keys or both (Table 1).

Dependent Variables

The primary dependent variable was the score on a 20-item multiple-choice and short-answer examination administered online at the end of the experimental treatment (Cronbach's $\alpha = .79$). Questions used on the posttest were extensively field tested.

The length of time (in minutes) from the start of the orientation module until the student completed the final examination was recorded by the computer. Time was used to assess the overall efficiency of different learners under the two treatments.

Results

Posttest Scores

A 2 $\times$ 2 analysis of variance (ANOVA) was conducted on posttest scores. There was a significant interaction between treatment and prior conceptual understanding, $F(1, 79) = 10.53$, $p < .001$, $MS_e = 37.81$. The results of the ANOVA indicated that there were significant differences between means. A means separation test (Duncan's method) showed that the mean posttest score of subjects with low prior conceptual understanding in the learner-controlled treatment was significantly different from the mean posttest scores of subjects in the other three treatments. There were no significant differences in mean posttest scores of subjects with high prior conceptual understanding (HPCU) in the program-controlled and learner-controlled treatments and subjects with low prior conceptual understanding (LPCU) in the program controlled condition. The means and standard deviations appear in Table 2.

Table 2
Means and Standard Deviations for Posttest Scores and Task Time

Condition	High prior understanding ($n = 20$)				Low prior understanding ($n = 20$)			
	Posttest		Task time		Posttest		Task time	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Learner-controlled	17.85	1.46	61.15	7.19	14.35	2.81	69.45	13.09
Program-controlled	18.00	1.12	66.35	8.21	17.25	1.74	71.85	9.26

Time on Task

The results of the ANOVA indicated that there was a significant interaction between treatment and prior conceptual understanding on task time, $F(1, 79) = 10.53, p < .01, MS_e = 1280.2$. The results of a Duncan means separation test indicated that HPCU subjects in the learner-controlled treatment were significantly more efficient than all other groups in their use of time. The means and standard deviations appear in Table 2.

Discussion

The results suggest that students' prior conceptual understanding of a particular content area is important in helping them understand the underlying organization and meaning of information presented. The organization and sequencing of information may markedly affect the students' performance (Eylon & Reif, 1984). The achievement differences between the low- and high-conceptual-understanding subjects on the posttest are consistent with Ausubel's (1968) theory of meaningful learning. Ausubel proposed that assimilation accounts for learning and forgetting and that retention is a direct effect of initial learning. He felt that the most important factor for learning is the prior presence of clear, stable, relevant ideas to act as anchors for the new material. Therefore, the prior conceptual structure of the subjects in this experiment may have acted as their advance organizer, providing the necessary anchors that would allow new material to be incorporated into the learner's cognitive structure. If the learner is familiar with the content, the presentation of well-structured hierarchical materials seems to be considerably less important.

The results of this experiment demonstrate that not all subjects were capable of making appropriate decisions. In terms of posttest scores, subjects performed better in the program-controlled treatment than in the learner-controlled treatment. After examining the computer records of strategies used by subjects during the experiment, it seemed that LPCU subjects tended to practice too little and to emphasize areas with which they already had some familiarity. They avoided unfamiliar or difficult topics. It seemed that LPCU subjects had difficulty determining how much practice they needed to learn a particular topic. LPCU subjects tended to disregard review and remediation options, repeat and overlearn topics that they already understood, and skip segments that they should have emphasized. In summary, LPCU subjects did not use good learning strategies and made poor sequencing decisions under the learner-controlled treatment.

On the other hand, HPCU subjects were able to function equally well in both treatments. Analysis of the computer records indicated that they selected fewer examples, emphasized topics that they did not understand, used practice questions to discover areas of weakness, chose remediation and review seg-

ments, and were able to effectively organize their learning strategies.

The results indicate that there is an interaction between learner control of instruction and prior familiarity with the material to be learned. Learners can be given more control if their prior understanding of a topic is relatively high; conversely, learners should be provided with more structure if their prior conceptual understanding of a topic is low. The results also suggest that students should be taught how to use control options more effectively. Efforts should continue to develop and test models for learning systems that have the capability to adapt to individual learners' needs.

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